

Foundation (Jalapeno)

Q1. Factor the trinomial $x^2 + 5x + 6$

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $(x - 2)(x - 3)$
- (D) $(x - 1)(x - 6)$
- (E) $(x + 2)(x - 3)$

Q2. Factor the trinomial $x^2 - 7x + 12$

- (A) $(x - 3)(x - 4)$
- (B) $(x - 1)(x - 12)$
- (C) $(x + 3)(x + 4)$
- (D) $(x + 1)(x + 12)$
- (E) $(x - 3)(x + 4)$

Q3. Factor the trinomial $x^2 + 2x - 15$

- (A) $(x + 5)(x - 3)$
- (B) $(x + 1)(x - 15)$
- (C) $(x - 5)(x + 3)$
- (D) $(x - 1)(x + 15)$
- (E) $(x + 5)(x + 3)$

Q4. Factor the trinomial $x^2 - 4x - 21$

- (A) $(x - 7)(x + 3)$
- (B) $(x - 1)(x - 21)$
- (C) $(x + 7)(x - 3)$
- (D) $(x + 1)(x + 21)$
- (E) $(x + 7)(x + 3)$

Q5. Factor the trinomial $2x^2 + 5x - 3$

- (A) $(2x - 1)(x + 3)$
- (B) $(2x + 1)(x - 3)$
- (C) $(2x - 1)(x - 3)$
- (D) $(2x + 1)(x + 3)$
- (E) $(x + 1)(2x - 3)$

Q6. Factor the trinomial $3x^2 + 2x - 1$

- (A) $(3x - 1)(x + 1)$
- (B) $(3x + 1)(x - 1)$
- (C) $(3x - 1)(x - 1)$
- (D) $(3x + 1)(x + 1)$
- (E) $(x + 1)(3x - 1)$

Q7. Factor the trinomial $2x^2 - x - 6$

- (A) $(2x + 3)(x - 2)$
- (B) $(2x - 3)(x + 2)$
- (C) $(x + 3)(2x - 2)$
- (D) $(x - 3)(2x + 2)$
- (E) $(2x + 2)(x - 3)$

Q8. Factor the trinomial $x^2 - 2x - 8$

- (A) $(x - 4)(x + 2)$
- (B) $(x - 2)(x + 4)$
- (C) $(x + 4)(x - 2)$
- (D) $(x - 1)(x - 8)$
- (E) $(x + 1)(x + 8)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 + 4x + 3$ and explain your steps.

Q10. Factor the trinomial $2x^2 - 5x - 3$ and show all your work.

Chef's Tips

- Encourage students to check their work by multiplying their factors.
- Remind students to use the FOIL method for factoring.
- Emphasize the importance of the correct signs when factoring.